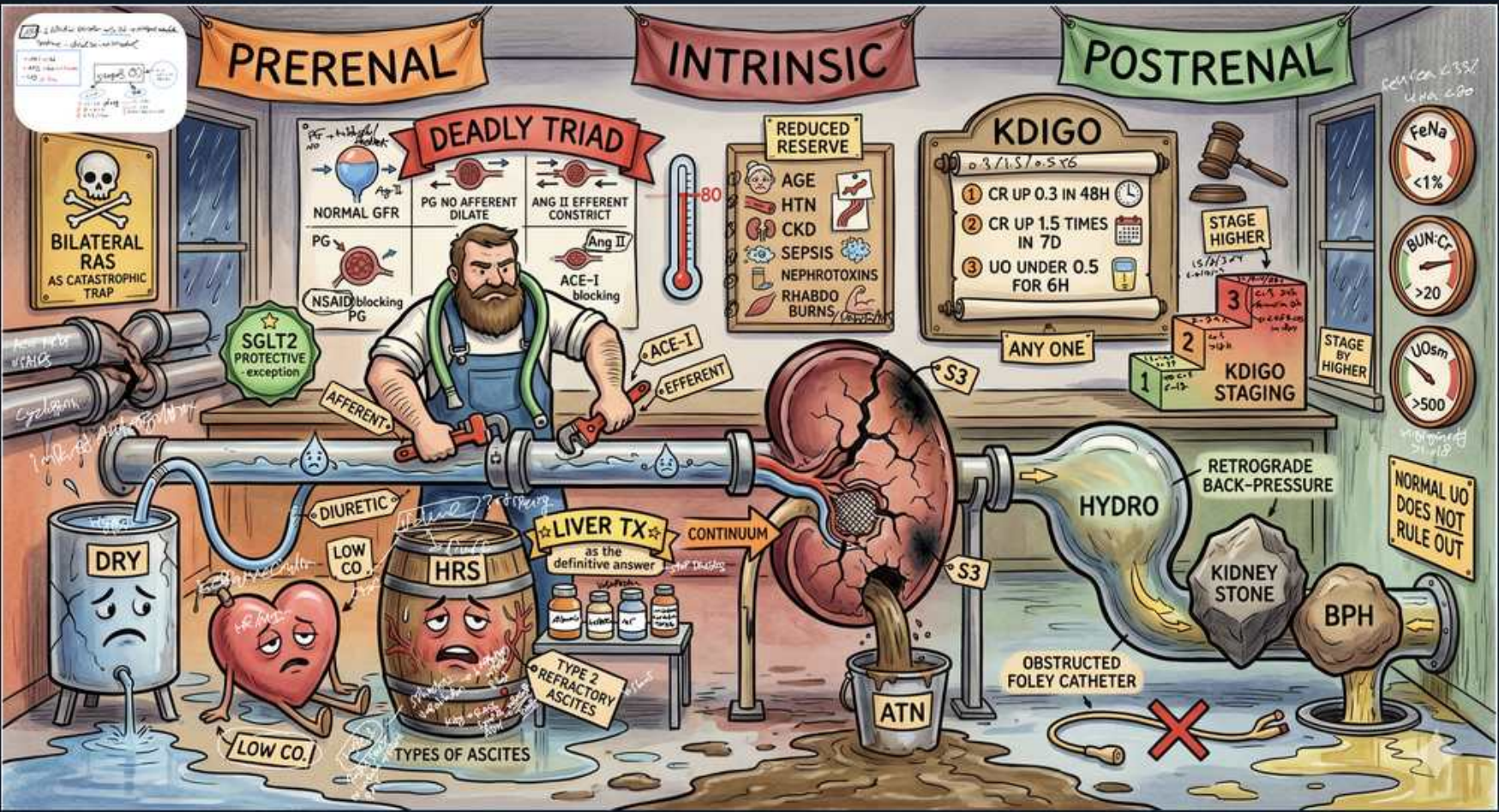


AKI — Overview & Prerenal



1. 3 categories

WHAT YOU SEE

PRERENAL / INTRINSIC / POSTRENAL banner across top

WHAT IT ENCODES

AKI is classified by the anatomic site of injury into prerenal (renal hypoperfusion, ~60–70% of cases), intrinsic/renal (parenchymal — ATN, AIN, GN, vascular), and postrenal (obstruction of urine outflow). This framework drives the entire workup: prerenal and postrenal are reversible if caught early, while intrinsic injury (especially ATN) takes days to weeks to recover. Always exclude the two reversible categories first.

BOARD TRAP

Treating the categories as mutually exclusive — sustained prerenal hypoperfusion progresses to intrinsic ATN, and a chronically obstructed kidney develops intrinsic damage, so a single patient can move along the prerenal→ATN continuum (the cracked-kidney 'continuum' in the sketch).

2. KDIGO criteria

WHAT YOU SEE

KDIGO box: Cr up 0.3 in 48h / Cr up 1.5x baseline / UO under 0.5 for 6h

WHAT IT ENCODES

AKI is diagnosed by ANY ONE of the KDIGO criteria: a rise in serum creatinine of ≥ 0.3 mg/dL within 48 hours, OR a rise to $\geq 1.5\times$ baseline known/presumed within 7 days, OR urine output < 0.5 mL/kg/h for ≥ 6 hours. Only one criterion is needed. These thresholds standardize early detection before overt failure.

BOARD TRAP

Thinking all three criteria must be met simultaneously — they are an 'any-of' definition, so a 0.3 mg/dL bump in creatinine over two days alone qualifies as AKI even with normal urine output.

3. KDIGO staging

WHAT YOU SEE

Stages 1-2-3, 'stage higher' arrows

WHAT IT ENCODES

KDIGO stages AKI 1–3 by severity: Stage 1 = Cr 1.5–1.9 $\times$ baseline or ≥ 0.3 rise (or UO < 0.5 mL/kg/h $\times 6$ –12h); Stage 2 = Cr 2.0–2.9 $\times$; Stage 3 = Cr $\geq 3.0\times$ baseline, Cr ≥ 4.0 mg/dL, initiation of dialysis, or anuria ≥ 12 h. Higher stage predicts worse mortality and progression to CKD.

BOARD TRAP

Forgetting that starting renal replacement therapy or anuria automatically makes it Stage 3 regardless of the creatinine value — staging is the worst of the creatinine and urine-output criteria, not just the number.

4. Prerenal = volume

WHAT YOU SEE

Dehydrated tap labeled DRY, LOW CO sad heart, flooded floor

WHAT IT ENCODES

Prerenal AKI is a functional drop in GFR from decreased renal perfusion — true volume depletion (vomiting, diarrhea, hemorrhage, burns, diuresis) or a low effective circulating volume (heart failure with low cardiac output, cirrhosis, sepsis, third-spacing). The tubules are structurally INTACT and respond appropriately by conserving sodium and water. GFR falls but the parenchyma is undamaged.

BOARD TRAP

Picking 'tubular cell death/necrosis is present' for prerenal — by definition prerenal tubules are structurally normal; cell death (muddy-brown casts) marks the transition to ATN once hypoperfusion is prolonged.

5. FENa <1%

WHAT YOU SEE

Gauge FENa <1

WHAT IT ENCODES

In prerenal AKI the fractional excretion of sodium is <1% because intact, RAAS-stimulated tubules avidly reabsorb sodium to defend volume. By contrast ATN gives FENa >2% from loss of reabsorptive capacity. FENa is invalid if the patient is on diuretics (use FEUrea <35% instead).

BOARD TRAP

Expecting a high FENa (>2%) in prerenal disease — that is the ATN pattern; prerenal characteristically gives FENa <1%, and reversing the two is a common exam mistake.

6. BUN:Cr >20

WHAT YOU SEE

Gauge BUN:Cr >20

WHAT IT ENCODES

A BUN:Cr ratio >20:1 supports prerenal azotemia: slow tubular flow and ADH-driven water reabsorption enhance passive urea reabsorption, while creatinine (not reabsorbed) does not rise proportionally. ATN, by contrast, gives a ratio near 10–15:1. The ratio is falsely raised by GI bleeding, steroids, and catabolic/high-protein states.

BOARD TRAP

Over-relying on a high BUN:Cr — a GI bleed raises BUN from absorbed blood protein and can mimic prerenal even when perfusion is fine, so corroborate with FENa, sediment, and clinical volume status.

7. Bland sediment

WHAT YOU SEE

Gauge UO >500 (concentrated urine)

WHAT IT ENCODES

Prerenal urine is concentrated — urine osmolality >500 mOsm/kg, specific gravity high, urine Na <20 mEq/L — reflecting intact tubular concentrating and Na-conserving function. The sediment is BLAND, containing at most hyaline casts and NO granular/cellular casts. This pattern (concentrated, low-Na, bland) is the prerenal urine fingerprint.

BOARD TRAP

Expecting muddy-brown granular casts in prerenal — those are the hallmark of ATN; finding them means intrinsic tubular injury, not prerenal physiology. Prerenal urine is bland.

8. Deadly triad

WHAT YOU SEE

DEADLY TRIAD: NSAID blocks PG (afferent), ACE-I blocks AngII (efferent), diuretic

WHAT IT ENCODES

The 'triple whammy' of an NSAID + an ACE-I/ARB + a diuretic precipitates prerenal AKI by simultaneously crippling renal autoregulation: the NSAID blocks afferent vasodilation, the ACE-I/ARB blocks efferent vasoconstriction, and the diuretic depletes volume. Each alone is usually tolerated; together they collapse GFR, especially in the elderly or volume-depleted.

BOARD TRAP

Stopping only one agent — the danger is the combination, and in an at-risk patient (CKD, dehydration, heart failure) the high-yield answer is to hold all three nephrotoxic/autoregulation-blocking drugs and restore volume.

9. Afferent dilate

WHAT YOU SEE

PG to AFFERENT dilate arrow, NSAID blocking

WHAT IT ENCODES

Prostaglandins (PGE2, PGI2) vasodilate the AFFERENT arteriole to maintain glomerular inflow and GFR during hypoperfusion. NSAIDs inhibit cyclooxygenase, removing this vasodilation, so the afferent constricts, inflow drops, and GFR falls — a prerenal AKI especially in volume-depleted or CKD patients who depend on PG-mediated autoregulation.

BOARD TRAP

Confusing which arteriole the NSAID hits — NSAIDs act on the AFFERENT (constricting it via loss of PG dilation); mixing this up with the efferent (the ACE-I site) is a frequent autoregulation question error.

10. Efferent constrict

WHAT YOU SEE

ANG II to EFFERENT constrict, ACE-I blocking

WHAT IT ENCODES

Angiotensin II preferentially constricts the EFFERENT arteriole, raising intraglomerular pressure to preserve GFR when perfusion is low. ACE-inhibitors and ARBs block this efferent tone, dropping intraglomerular pressure and GFR — beneficial long-term (renoprotective, reduces proteinuria) but harmful acutely in hypoperfusion or bilateral renal artery stenosis.

BOARD TRAP

Reading an expected creatinine bump (up to ~30%) after starting an ACE-I as 'AKI requiring drug cessation' — a rise $\leq 30\%$ reflects intended efferent dilation and is acceptable; only a $>30\%$ rise (or hyperkalemia) warrants stopping and evaluating for bilateral RAS.

11. Bilateral RAS

WHAT YOU SEE

Skull/poison sign BILATERAL RAS as catastrophic trap

WHAT IT ENCODES

In bilateral renal artery stenosis (or stenosis of a solitary functioning kidney), GFR is critically dependent on Ang-II-mediated efferent constriction. Starting an ACE-I/ARB removes that efferent tone and causes catastrophic, acute drop in GFR. A steep creatinine rise after starting an ACE-I should trigger evaluation for bilateral RAS (often older atherosclerotic patients or flash pulmonary edema).

BOARD TRAP

Continuing or up-titrating the ACE-I after creatinine spikes — the correct response is to stop the ACE-I/ARB and investigate for bilateral RAS; pushing on can cause irreversible loss of function.

12. HRS

WHAT YOU SEE

HRS barrel, liver Tx as definitive answer, ascites

WHAT IT ENCODES

Hepatorenal syndrome is a FUNCTIONAL prerenal AKI in advanced cirrhosis: splanchnic vasodilation lowers effective arterial volume, triggering intense renal vasoconstriction. It is a diagnosis of exclusion (no improvement after stopping diuretics and giving albumin, bland sediment, urine Na <10 , no shock/nephrotoxins). Medical bridge is albumin + vasoconstrictors (terlipressin or octreotide+midodrine+albumin); LIVER TRANSPLANT is definitive.

BOARD TRAP

Diagnosing ATN because there is AKI in a sick cirrhotic — HRS has BLAND sediment and very low urine Na (kidneys are structurally normal), whereas ATN shows muddy-brown casts and FENa $>2\%$; the discriminator is the urine, and the key first step is a volume challenge with albumin.

13. Responds to fluids

WHAT YOU SEE

The DRY tap and sad LOW-CO heart at the base feeding the kidney once the floor is re-flooded — restoring volume/perfusion is what reverses prerenal AKI, the defining feature that the dry source is simply refilled.

WHAT IT ENCODES

The defining feature of prerenal azotemia is rapid reversibility: restoring perfusion (IV fluids in hypovolemia, or improving cardiac output/treating the underlying cause) returns creatinine toward baseline within 24–72 hours. Persistent azotemia despite adequate volume repletion signals that prerenal injury has progressed to intrinsic ATN.

BOARD TRAP

Continuing to bolus fluids when creatinine fails to improve and the patient is now euvolemic/overloaded — failure to respond means ATN (or another intrinsic cause), and pouring in more fluid causes pulmonary edema rather than reversing the AKI.

14. Postrenal obstruction

WHAT YOU SEE

HYDRO kidney, KIDNEY STONE, BPH, obstructed Foley, retrograde back-pressure

WHAT IT ENCODES

Postrenal AKI requires obstruction at a level affecting BOTH kidneys (or a solitary kidney): bladder outlet obstruction from BPH/prostate cancer, bilateral ureteral stones or strictures, retroperitoneal fibrosis/malignancy, or a blocked/clamped Foley catheter. Retrograde back-pressure transmits to the tubules, raises intratubular pressure, and drops net filtration; prolonged obstruction causes hydronephrosis and eventual intrinsic damage. Renal ultrasound (hydronephrosis) and relief of obstruction (Foley, nephrostomy, stent) are diagnostic and therapeutic.

BOARD TRAP

Excluding postrenal because the patient is making urine — a single unilateral stone with a healthy contralateral kidney rarely raises creatinine, and a BLOCKED Foley is a classic missed cause; the discriminator is bedside bladder scan/flushing the catheter and ultrasound for bilateral hydronephrosis.

15. Post-obstructive diuresis

WHAT YOU SEE

Polyuria gushing out after the obstruction downstream is decompressed — the flood released once the blockage clears IS the post-obstructive diuresis to monitor and partly replace.

WHAT IT ENCODES

After relief of bilateral obstruction, patients can develop post-obstructive diuresis — a transient massive polyuria from retained urea/solute (appropriate) and impaired tubular concentrating/Na handling (inappropriate). Monitor urine output, volume status, and electrolytes closely; replace roughly half to two-thirds of urinary losses to avoid hypovolemia while allowing the kidney to offload retained solute.

BOARD TRAP

Aggressively matching every milliliter of urine output one-for-one — over-replacement perpetuates the diuresis and causes ongoing volume/electrolyte derangement; the correct approach is partial, monitored replacement and watching for the diuresis to self-limit.

16. Reduced renal reserve

WHAT YOU SEE

REDUCED RESERVE box: AGE, HTN, CKD, sepsis, nephrotoxins

WHAT IT ENCODES

Reduced renal reserve makes a kidney far more vulnerable to AKI from any insult: advanced age, hypertension, diabetes, CKD, and recent sepsis or nephrotoxin exposure all lower the threshold at which hypoperfusion or a drug tips into injury. These patients lose autoregulatory cushion, so a minor volume loss or a single NSAID can precipitate AKI.

BOARD TRAP

Assuming a 'normal-sized creatinine' patient has normal reserve — an elderly or CKD patient with low muscle mass can have a 'normal' creatinine yet markedly reduced GFR and reserve, so they tolerate nephrotoxins and volume loss poorly.